Chapter 19 Notes

Section 2

Notes

For the model in this section to work, we need $\omega_2 \in M$. However, it is not clear to my why this is.

If we assume that $\omega_2 \in M$, here is why $\langle \mathcal{G}, \mathcal{R} \rangle$ is an M-set. It is easy to see that each particular forcing condition is in M - each particular one can be explicitly defined, as it is assumed that $p(\alpha) = \langle \emptyset, \emptyset \rangle$ for all but a finite number of $\alpha \in \omega_2$. The set $\mathcal{G}$ of all forcing conditions is a subset of $\mathcal{P}(\mathcal{P}(\omega_2 \times (\mathcal{P}(\omega \times \omega) \cap M)) \cap M) \cap M$, which is in M. $\mathcal{G}$ is also easily seen to be definable over M, and consequently, $\mathcal{G} \in M$. Then $\mathcal{R}$ is also easily seen to be in M, as it is definable over M, and is a subset of $\mathcal{G} \times \mathcal{G}$, which is in M (as we have shown $\mathcal{G}$ to be in M).

Exercise 2.1

This is basically the same proof as that of Theorem 4.1 of Chapter 18.

Proof. We prove that, for each $\alpha \in \omega_2$, $[\![\sim\text{Const}(f_\alpha)]\!]$ is valid in the model M. Suppose, to the contrary, that $p \nvDash [\![\sim\text{Const}(f_\alpha)]\!]$ for some p. Then for some p' with $p\mathcal{R}p'$, $p' \vDash [\![\text{Const}(f_\alpha)]\!]$. By Theorem 3.2 of Chapter 18, it follows that $p \vDash \Diamond [\![f_\alpha \approx \hat{c}]\!]$ for some contrsuctible set $c \in M$. This means that, for some r with $p'\mathcal{R}r$, $r \vDash [\![f_\alpha \approx \hat{c}]\!]$. We derive a contradiction from this. r is some forcing condition, so suppose that $r(\alpha) = \langle P, N \rangle$. There are then two cases.

Case 1) $c \subseteq P$. Because both P and N are finite, we can pick some integer n such that $n \notin P \cup N$ (thus also $n \notin c$). Let $P' = P \cup \{n\}$. Then there is some forcing condition q such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = r(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P', N \rangle$. Obviously $r\mathcal{R}q$. Because $c \in P'$, it follows that $q \vDash [\![\hat{n} \in f_\alpha]\!]$. But because $n \notin c$, $q \vDash [\![\sim(\hat{n} \in \hat{c})]\!]$. These two things jointly imply $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$, as $r \vDash [\![f_\alpha \approx \hat{c}]\!]$ and $r\mathcal{R}q$. Thus we have our contradiction in this case.

Case 2) $c \nsubseteq P$. Pick some n in c but not in P. Let $N' = N \cup \{n\}$. Then there is some forcing condition q such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = r(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P, N' \rangle$. Obviously $r\mathcal{R}q$. Because $n \in c$, it follows that $q \vDash [\![\hat{n} \in \hat{c}]\!]$. But because $n \in N'$, it follows that $q \vDash [\![\sim(\hat{n} \in f_\alpha)]\!]$ (Reason: Suppose, for some q' with $q\mathcal{R}q'$, that $q' \vDash \hat{n} \in f_\alpha$. Then for some x, $q' \vDash [\![x \epsilon f_\alpha]\!]$ and $q' \vDash [\![x \approx \hat{n}]\!]$. This means x is $\hat{a}$ for some a, and thus, by Lemma 3.2 of Chapter 17, a and n are the same set. But this means $q' \vDash [\![\hat{n} \epsilon f_\alpha]\!]$, which cannot happen, as $n \in N'$). These two things imply that $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$ (for the same reason as in the first case), which is our contradiction.

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